

Chapter - IV Information Gathering

→ Measurement scales for questionnaire

- ① Nominal
 - ② Interval
- descriptions

→ Problems with Scale

- ① Leniency
- ② Central Tendency
- ③ Halo Effect

① Leniency :

"How was your experience?"

- ① Poor
 - ② Fair
 - ③ Average
 - ④ Good
 - ⑤ Excellent
- Easy Raters
- arises leniency problem

Caused by easy raters :

Solution is to move the "average" category to the left or right of center

② Central Tendency :

<<central biasness>> ③ Average

- Solⁿ
- make the distance smaller at the two ends
 - adjust strength of descriptors
 - a scale with more points

- ① Slightly below avg
- ② Below avg
- ③ Avg
- ④ Above average
- ⑤ Slightly above avg

③ Halo Effect

→ when the impression formed in one question carries into the next question

→ solⁿ: place one trait and several items on each page

→ Designing Questionnaire

→ Order of questions

↳ important first,

↳ cluster items of similar content together.

→ Question Types

↳ Closed

↳ Open Ended

Solve: বইয়ের last question গুলো

Using Data Flow Diagrams

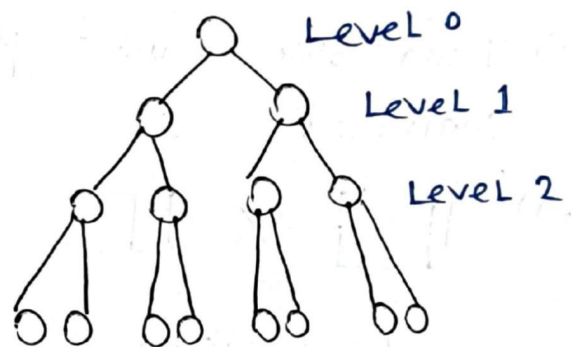
Context Level DFD << mid >>

Other DFDs depend on Context Level DFD for accuracy.

Context Level (Level 0)

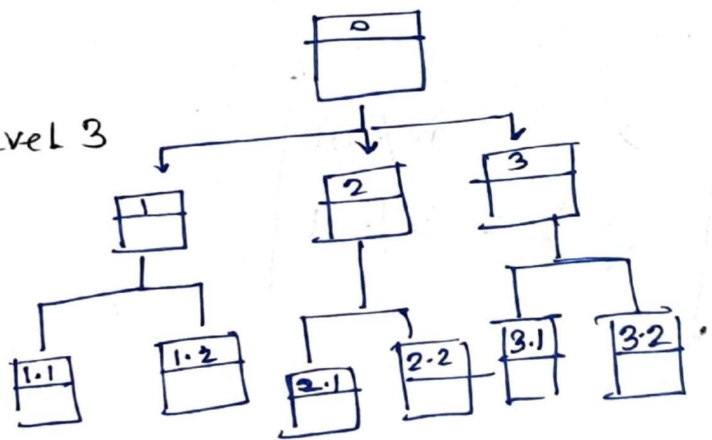
Level 1 → max 9 process
Level 2
Level n

Top down approach
Equivalent of trees



Primitive process: Those processes that cannot be exploded.

Level 0, Level 1, Level 2, Level 3
required for
this course

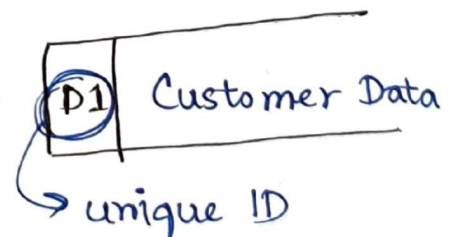


Required Symbols (4):



open ended rectangle
'Data Store'

Data Store



DFD Rules :

Shayan

① Cannot connect two entities



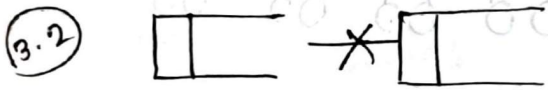
② Always ~~process~~ ^{প্রসেস} process ~~in & out~~ ^{in & out থাকবে} in & out থাকবে



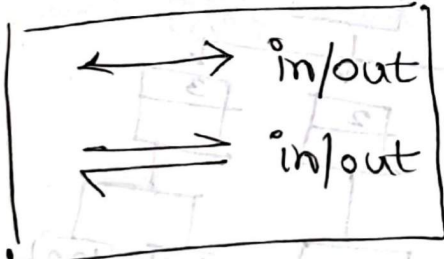
③ Data store always process-র সাথে connected.



NOT Entity - Data Store

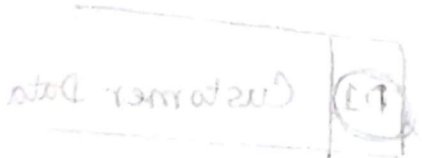


NOT DataStore - DataStore



Level 0, Level 1, Level 2, Level 3

Required Symbols (প্রয়োজনীয় সিম্বলস)



Customer ID

Chapter - 7

Context Level / Level - 0



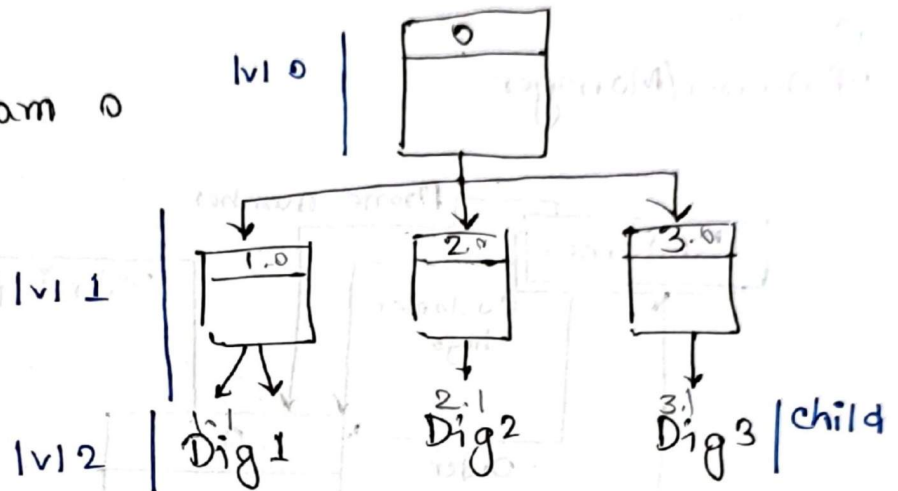
Level 1 DFD / Diagram 0



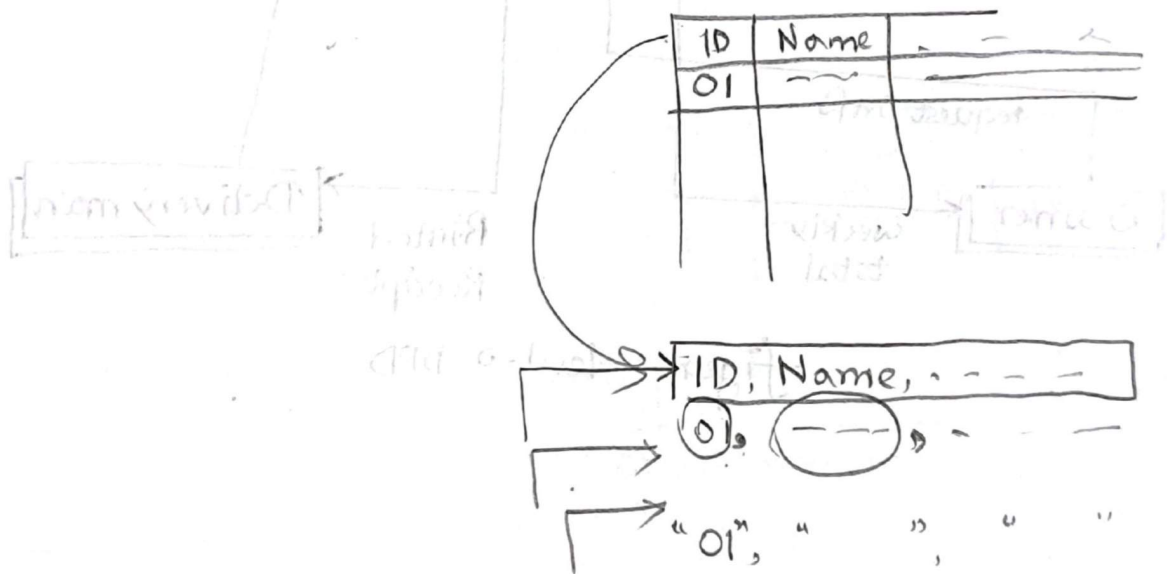
<child>



Level 2



A child diagram cannot produce output or receive input that has not being produced or received by parent.



PERFECT PIZZA

Context Level (Level 0) DFD

Entities:

- ① Customer
- ② Delivery man
- ③ Cook
- ④ Owner/Manager

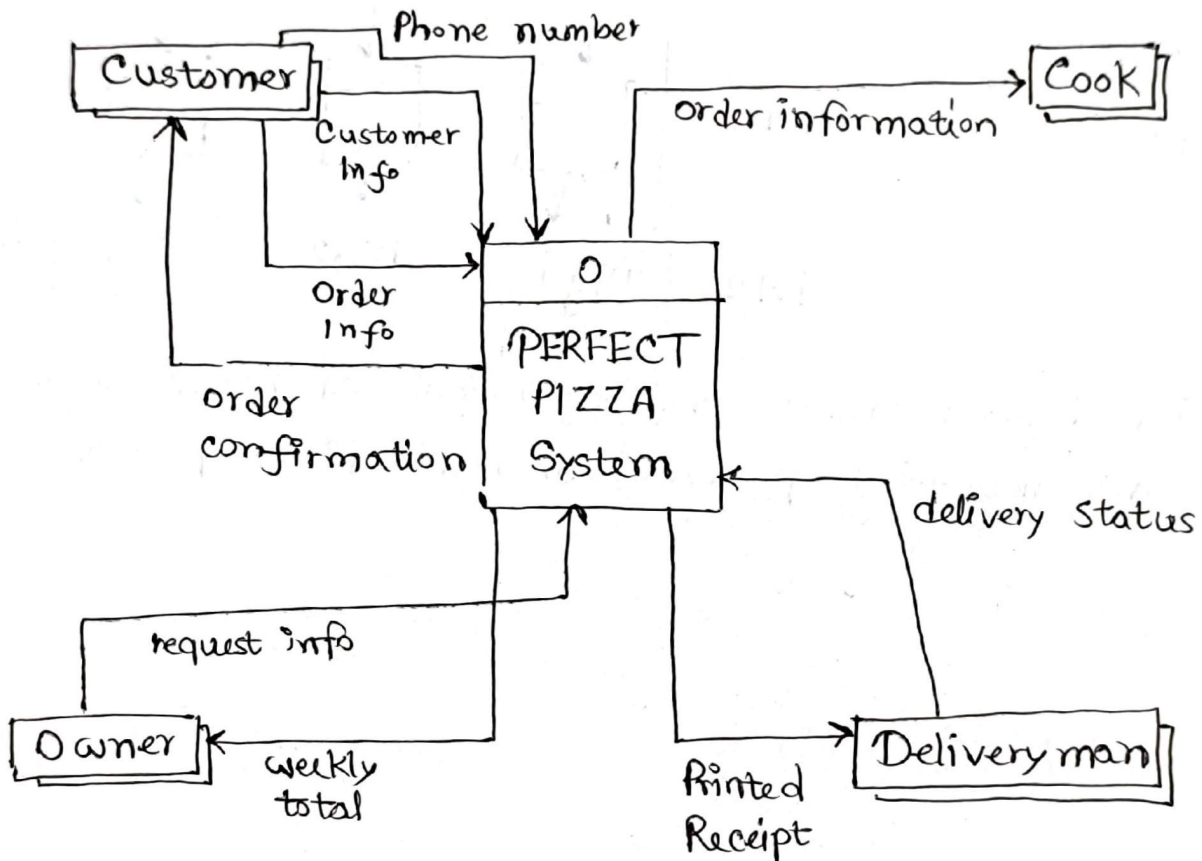


figure: level-0 DFD

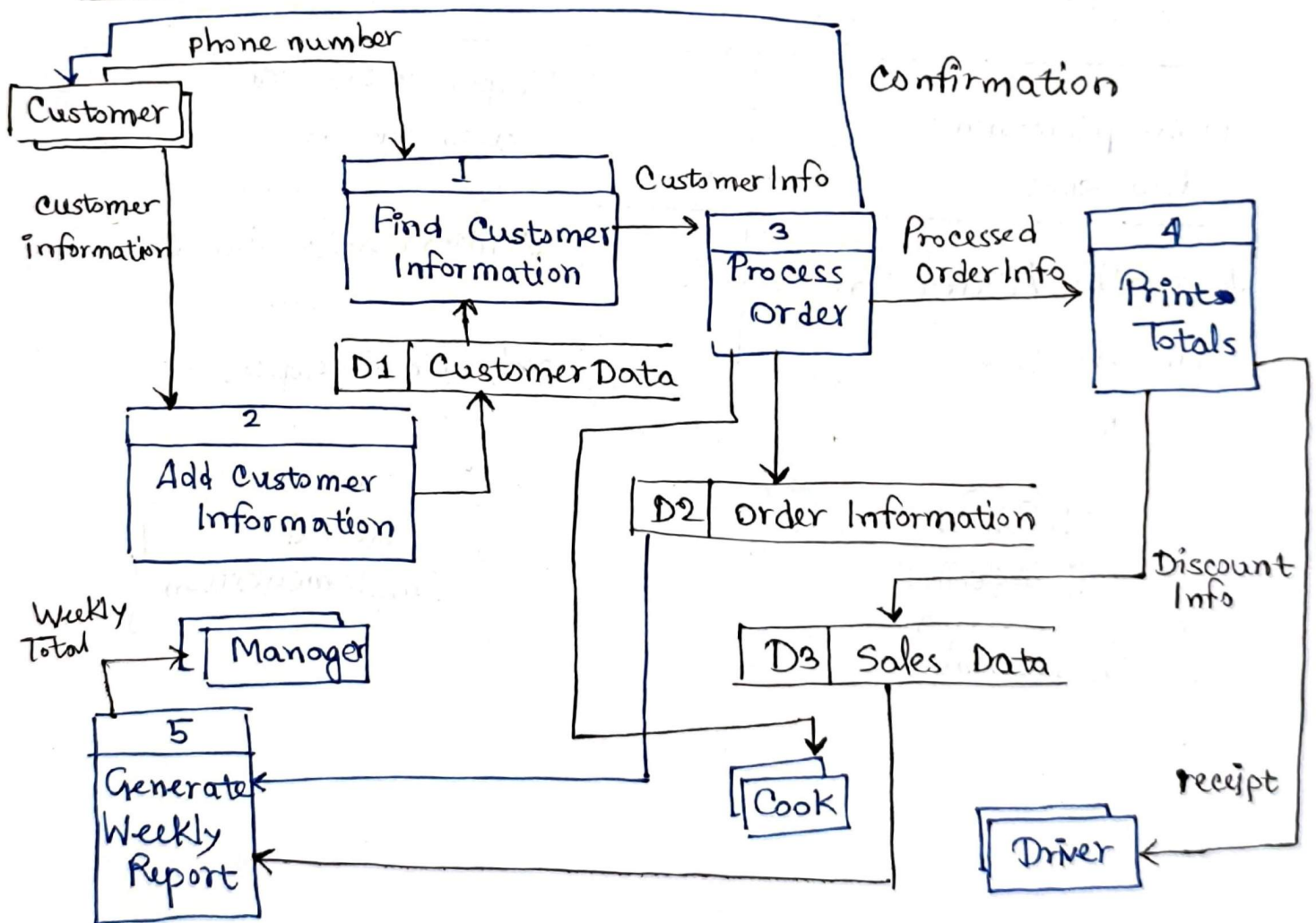
Level 1 DFD

Processes :

- ① Find customer information
- ② Add customer information
- ③ Process order
- ④ Print total
- ⑤ Generate weekly report

Data Stores :

- [D1] Customer Data
- [D2] Order Information
- [D3] Sales Data



#self study : Level 2 → Print totals : Discount Receipt

Logical DFD & Physical DFD

Logical DFD

Shows the logic/business rules of the system.

No manual processes are shown.

Logical name of the components

Shows business activities

Only permanent data store

Verification/Validation are not shown

প্রত্যক্ষ
মা কল্পনাম
theoretical

Physical DFD

Shows the actual representation of the system.

Manual processes are shown.

Use actual name of the component.

Shows the modules/programs

Temporary ones are also shown

Verification/Validation are shown.

only for developers

actual implementation